CR-001: Dedicated US stack on a second cloud (US-West primary, US-East failover)

Status: DESIGN — not provisioned, released or tested. Per the brief, this CR was issued 24 hours into the assignment and is answered here as a design.

Release gate. Implementation starts only after CAB approval of this CR. Go-live is **no earlier than 24 h after approval**, and only once the failover game day below has passed and the tenant owner has signed off.

Change summary. One dedicated US tenant on **Google Cloud**, a different provider from the AWS EU fleet. This removes the single-provider dependency for that tenant and adds regional failover:

- primary in **us-west1**, warm standby in **us-east4**
- US-resident data; EU tenant data is never replicated to it

Design: same platform, new provider module.

- **Provider layer:** add `infra/modules/gcp-stack`, exporting the existing output contract (`cluster_endpoint`, `postgres_*`, `redis_*`, `secret_store`, `region`, `data_residency=us`). The residency validation allows `us-*` regions only.
- **Service mapping:** EKS → private regional GKE · RDS → Cloud SQL PostgreSQL HA plus a cross-region replica · ElastiCache → Memorystore Redis Standard · Secrets Manager → Secret Manager + ESO · KMS → Cloud KMS · IRSA → Workload Identity · ALB/Route 53 → global HTTPS LB/Cloud DNS.
- **CI:** GitHub OIDC into GCP Workload Identity Federation (no keys).
- **Platform and app layers:** unchanged, with the same chart and the **same signed digest** from GHCR.
- **Fleet:** added as **ring 3**, promoted last.
- **Topology** (grey "CR-001" box in the `architecture.md` diagram): users → GCP global LB and Cloud DNS, with health checks from 3 locations → **us-west1** primary (GKE, Cloud SQL HA, Memorystore), plus async replication to the **us-east4** warm standby (minimum-size GKE, read replica). Argo CD syncs both regions from the same Git and signed digest.

Replication, RPO and RTO.

Component	Mode	RPO	RTO
PostgreSQL	Cloud SQL HA (zonal) + **async** cross-region replica	≤ 1 min (alert if lag > 30 s)	≤ 15 min (promote replica)
Redis queue	**Not replicated.** Pre-created standby; job intents also written to a PostgreSQL **outbox** (an app change)	0 with outbox (else in-flight jobs lost)	≤ 15 min
GKE workloads	Standby at minimum size, synced by Argo CD from the same Git/digest	n/a (stateless)	≤ 10 min scale-up
Backups	PITR 7 d + daily export to a US-only bucket	≤ 5 min	1–2 h

Tenant target: RPO ≤ 1 min, RTO ≤ 30 min. A regional failover exceeds the 21.9-minute monthly budget of 99.95%; that is accepted for region loss only.

Failover.

- **Trigger:** all three conditions for 5 minutes: external probes from 3 locations failing, burn-rate page firing, provider regional incident. A **human approves**, because database split-brain is worse than a few minutes' delay.
- **Steps:** (1) freeze deploys; (2) promote the replica; (3) repoint secrets; (4) scale us-east4; (5) re-enqueue the outbox; (6) switch LB/DNS (TTL 60 s); (7) verify SLIs.
- **Failback:** rebuild a replica in us-west1, wait for lag under 5 s, switch over in a maintenance window. At-least-once delivery plus idempotency markers make re-enqueued jobs safe.

Network, security, observability.

- **Network:** private GKE per region, Private Service Connect to the data services, HA VPN for replication only, **no path to EU stacks**, Cloud Armor.
- **Isolation:** a separate GCP project and US state bucket.
- **Monitoring:** same alert rules and `tenant` labels, with US telemetry kept in the US; new alerts on replica lag, standby readiness and outbox depth.

Cost (GCP list prices, approximate, unverified): primary ≈ $1,360/month (GKE, 3× n2-standard-4, Cloud SQL HA 4 vCPU/16 GB, Memorystore 5 GB, LB/NAT/logs) + warm standby ≈ $680 + replication egress ≈ $50 = **≈ $2.1k/month**. That takes the fleet to ≈ $10.3k steady, **over the $9k budget** unless offset (Savings Plans, Graviton, CDN) or charged to the tenant.

Plan (about 2 weeks): (1) GCP project, state bucket and WIF; (2) `gcp-stack` module and `live/gcp/dedicated-us`, with network and firewall; (3) API outbox; (4) platform layer with `target=gcp`; (5) Argo CD registration, ring 3 and data sync; (6) health validation, then a failover **and** failback game day with measured RPO/RTO; (7) residency audit. **Rollback of the change:** before cut-over, don't switch DNS; keep the tenant on its current serving path and `terraform destroy` the GCP stack (no data has moved). After cut-over, use failback. **After release:** a 7-day hypercare with replica-lag, standby and outbox alerts.

Risks → mitigations: *cross-cloud networking* → HA VPN for replication only, IaC-only firewall, egress budget alert; *data consistency and replication lag* → alert at 30 s, fail over only after checking lag, outbox for jobs; *DNS propagation* → 60 s TTL, LB health checks steer first; *failover automation fails* → human-approved runbook, quarterly game day, warm standby; *provider dependency and behaviour differences* → same output contract, GCP-specific runbook, measured in the game day; *cost* (≈ $2.1k per month, over budget) → charge to the tenant or offset, re-baseline after 30 days.